

Chapter 111

Miscellaneous Nodes

This chapter details miscellaneous nodes within Fusion.

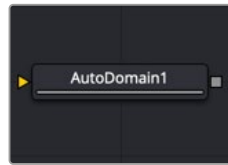
The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Auto Domain [ADoD]



The Auto Domain node

Auto Domain Node Introduction

The Auto Domain node automatically sets the image's domain of definition (DoD) based on bounds of the input image's background Canvas color. It does not change the image's physical dimensions. Some EXR images come with optimized DoDs already set, but other formats do not. For formats other than EXR, this node can speed up compositions by optimizing the DoD based on the content rather than the frame's dimensions.

For example, a CG character rarely takes up the entire frame of an image. With this type of image, the Auto Domain node sets the DoD to a rectangular region by comparing image pixels with the Canvas color. The Canvas color indicates what color the pixels are outside the DoD. By default, unless a Canvas color is set using the Set Canvas Color node, the color is set to black. This default works well when an image has a premultiplied alpha channel. The result is a DoD that encompasses the portion of the clip that contains only the character. The DoD is updated on each frame to accommodate changes, such as a character walking closer to the camera. However, if a clip does not contain an alpha channel, the Set Canvas Color node can be used to define the Canvas color as solid alpha with a color that matches the solid background.

For more detail on the Set Canvas Color node, see *Chapter 94, "Color Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 34 in the Fusion Reference Manual.

NOTE: The Domain of Definition is a bounding box that encompasses pixels that have a nonzero value. The DoD is used to limit image-processing calculations and speeds up rendering.

Inputs

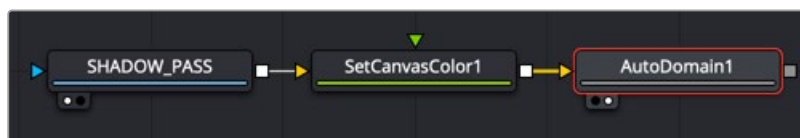
The single input on the Auto Domain node is used to connect a 2D image and an effect mask, which can be used to limit the blurred area.

Input: The orange input is used for the primary 2D image that is blurred.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the blur to only those pixels within the mask.

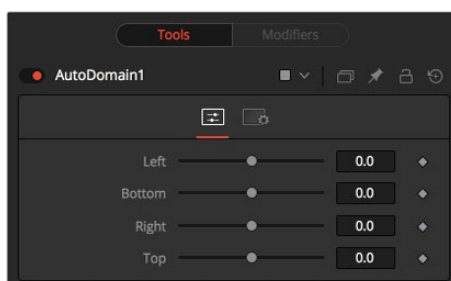
Basic Node Setup

The example below assumes that the image does not contain an alpha channel but a CG character rendered against a black background. This is common for different render passes like specular or shadows, for example. The image is connected to a Set Canvas Color node, which sets the Canvas color to black with a solid alpha. The Set Canvas Color then connects to the Auto Domain node, which detects the pixels and sets the DoD. If the original image contained a premultiplied alpha channel, the Set Canvas Color would not be needed, and the image could be connected directly into the Auto Domain node.



An Auto Domain node automatically limits the area of image processing.

Inspector



The Auto Domain Controls tab

Controls Tab

In most cases, the Auto Domain node automatically calculates the DoD bounding box; however, the rectangular shape can be modified using the Controls tab in the Inspector.

Left

Defines the left border of the search area of the ADoD. Higher values on this slider move the left border toward the right, excluding more data from the left margin.

1 represents the right border of the image; 0 represents the left border. The slider defaults to 0 (left border).

Bottom

Defines the bottom border of the search area of the ADoD. Higher values on this slider move the bottom border toward the top, excluding more data from the bottom margin.

1 represents the top border of the image; 0 represents the bottom border. The slider defaults to 0 (bottom border).

Right

Defines the right border of the search area of the ADoD. Higher values on this slider move the right border toward the left, excluding more data from the right margin.

1 represents the right border of the image; 0 represents the left border. The slider defaults to 1 (right border).

Top

Defines the top border of the search area of the ADoD. Higher values on this slider move the top border toward the bottom, excluding more data from the top margin.

1 represents the top border of the image; 0 represents the bottom border. The slider defaults to 1 (top border).

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Change Depth [CD]



The Change Depth node

Change Depth Node Introduction

The Change Depth node has one simple use, and that is to change the bits per color channel used to process a node. This node is often used after color correcting 32-bit floating-point image files, converting them from float processing to 16-bit per channel to preserve memory and performance.

It can also be useful if, from a certain point in your node tree, you feel the need to process your images in a higher bit depth than their original one or to reduce the bit depth to save memory.

Inputs

The single input on the Change Depth node is used to connect a 2D image and an effect mask, which can be used to limit the blurred area.

Input: The orange input is used for the primary 2D image to be converted.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes or bitmaps from other tools. Connecting a mask to this input limits the blur to only those pixels within the mask.

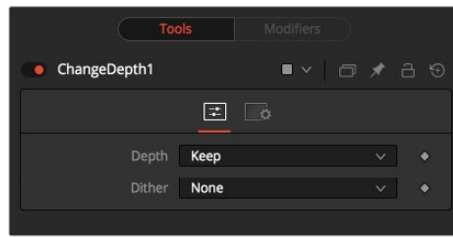
Basic Node Setup

Below, a Loader node in Fusion Studio is color corrected and then down converted from a floating-point image to a 16-bit image to save image-processing time and memory.



A Change Depth node placed after color correction is done on a floating-point image.

Inspector



The Change Depth Controls tab

Controls Tab

The two controls for this node are the Depth menu and the Dither menu. These two menus are used to convert and adjust the color depth of the image.

Depth

The Keep setting doesn't do anything to the image but instead keeps the input depth. The other options change the bit depth of the image to the respective value.

Dither

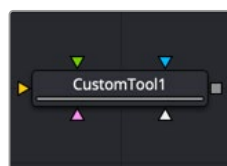
When down converting from a higher bit depth, it can be useful to add Error Diffusion or Additive Noise to camouflage artifacts that result from problematic (high-contrast) areas.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Custom Tool [CT]



The Custom Tool node

Custom Tool Node Introduction

The Custom Tool node is the most complex and the most powerful node in Fusion. It is used to create custom expressions and filters to modify an image. In addition to providing three image inputs, the Custom Tool node allows for the connection of up to eight numeric inputs and as many as four XY position values from other controls and parameters in the node tree.

Per-pixel calculations can be performed on the Red, Green, Blue, Alpha, Z, Z-Coverage, UV texture coords, XYZ Normals, RGBA background color, and XY motion vector channels of the images.

You should be moderately experienced with scripting, or C++ programming, to understand the structure and terminology used by the Custom Tool node.

Inputs

The Custom Tool node has three image inputs, a matte input, and an effect mask input.

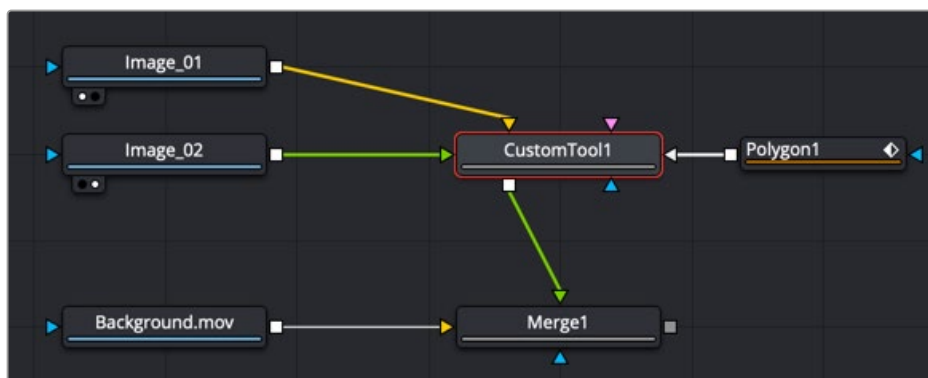
Input: The orange, green, and magenta inputs combine 2D images to make your composite. When entering them into the Custom Tool fields, they are referred to as c1, c2 and c3 (c standard for all three R, G, B channels)

Matte Input: The white input is for a matte created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a matte to this input allows a matte to be combined into any equation. When entering the matte into the Custom Tool fields, it is referred to as m1.

Effect Mask: The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Custom Tool effect to only those pixels within the mask.

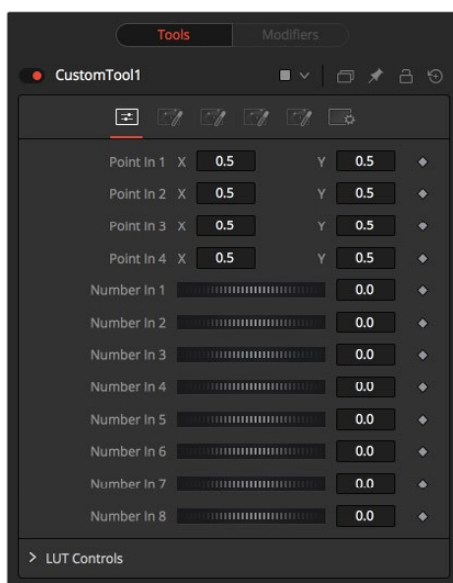
Basic Node Setup

The Custom Tool below takes two image inputs and a matte input, and then combines them using some calculation. The result can be output to a Merge or other image-processing nodes.



A Custom Tool is used to build your own effects using C++ and scripting.

Inspector



The Custom Tool Controls tab

Controls Tab

Point in 1-4, X and Y

These four controls are 2D X and Y center controls that are available to expressions entered in the Setup, Intermediate, and Channels tabs as variables `p1x`, `p1y`, ..., `p4x`, `p4y`. They are normal positional controls and can be animated or connected to modifiers as any other node might.

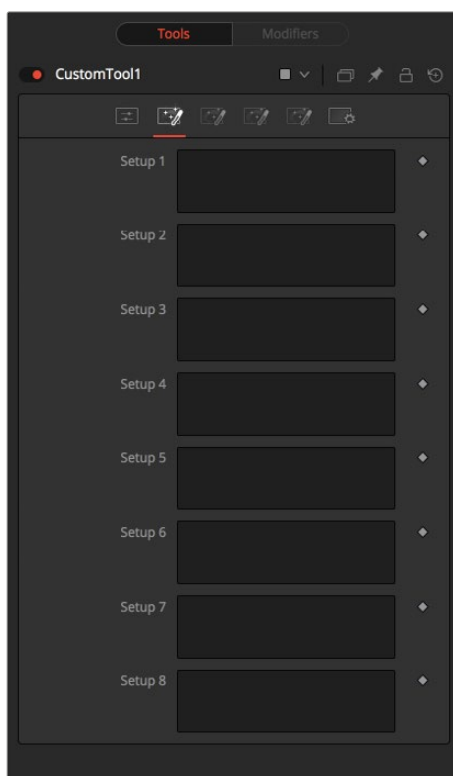
Number in 1-8

The values of these controls are available to expressions entered in the Setup, Intermediate, and Channels tabs as variables `n1`, `n2`, `n3`, ..., `n8`. They are normal slider controls and can be animated or connected to modifiers exactly as any other node might.

LUT in 1-4

The Custom Tool node provides 4 LUT splines. The values of these controls are available to expressions entered in the Setup, Intermediate, and Channels tabs using the `getlut#` function. For example, setting the R, G, B, and A expressions to `getlut1(r1)`, `getlut2(g1)`, `getlut3(b1)`, and `getlut4(a1)`, respectively, would cause the Custom Tool node to mimic the Color Curves node.

These controls can be renamed using the options in the Config tab to make their meanings more apparent, but expressions still see the values as `n1`, `n2`, ..., `n8`.



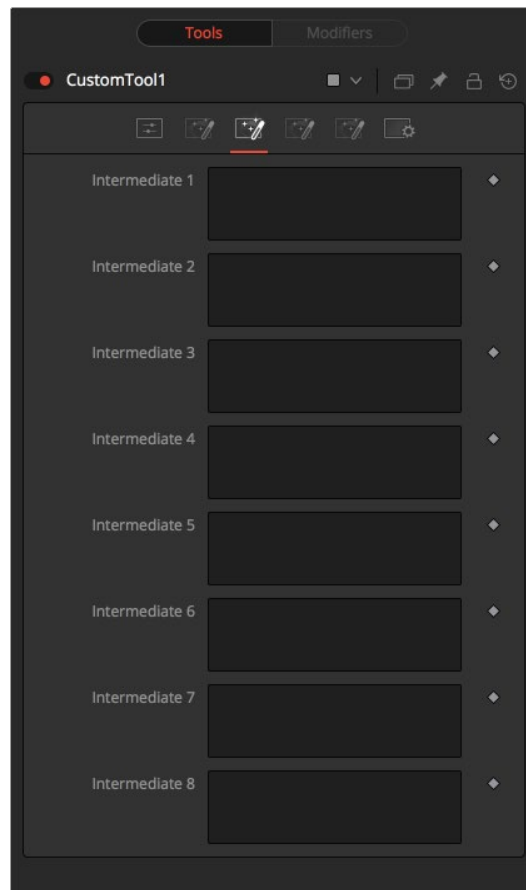
The Custom Tool Setup tab

Custom Tool Setup Tab

Setup 1-4

Up to four separate expressions can be calculated in the Setup tab of the Custom Tool node. The Setup expressions are evaluated once per frame before any other calculations are performed. The results are then made available to the other expressions in the Custom Tool node as variables `s1`, `s2`, `s3`, and `s4`.

NOTE: Because these expressions are evaluated once per frame only and not for each pixel, it makes no sense to use per-pixel variables like X and Y or channel variables like r1, g1, b1. Allowable values include constants, variables such as n1..n8, time, W and H, and functions like sin() or getr1d().

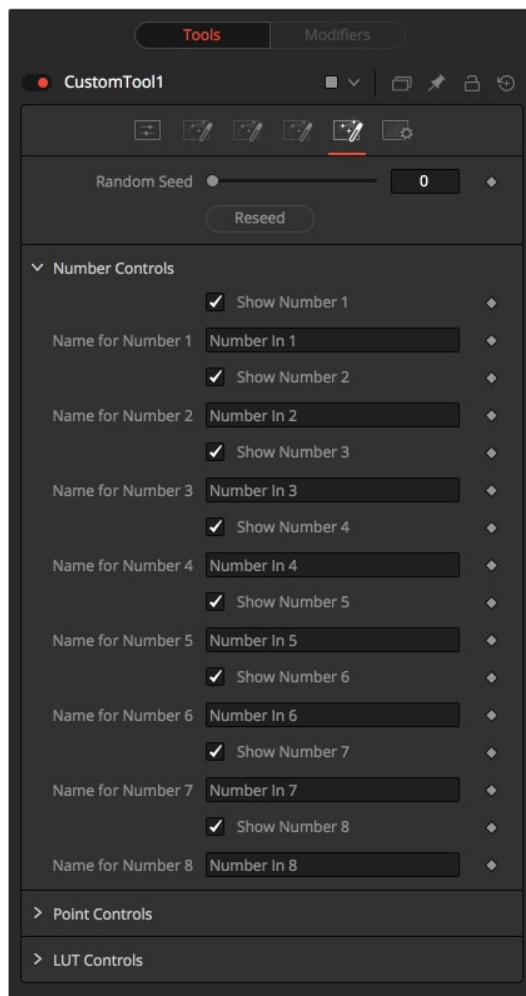


The Custom Tool Intermediate tab

Custom Tool Intermediate Tab

Intermediate 1-4

An additional four expressions can be calculated in the Inter tab. The Inter expressions are evaluated once per pixel after the Setup expressions are evaluated but before the Channel expressions are evaluated. Per-pixel channel variables like r1, g1, b1, and a1 are allowable. Results are available as variables i1, i2, i3, and i4.



The Custom Tool Config tab

Custom Tool Config Tab

Random Seed

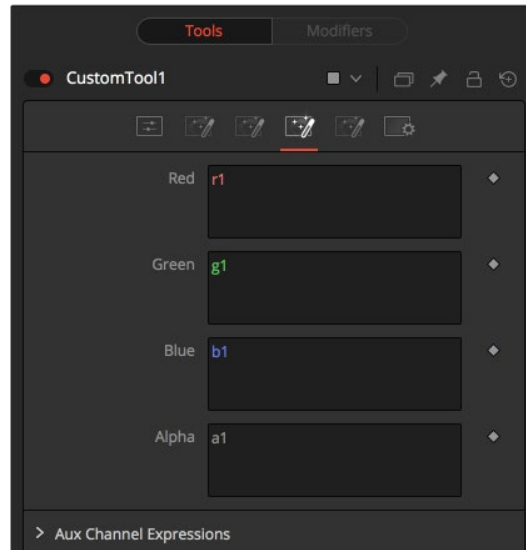
Use this to set the seed for the `rand()` and `rand8()` functions. Click the Randomize button to set the seed to a random value. This control may be needed if multiple Custom Tool nodes are required with different random results for each.

Number Controls

There are eight sets of Number controls, corresponding to the eight Number In sliders in the Controls tab. Uncheck the Show Number checkbox to hide the corresponding Number In slider, or edit the Name for Number text field to change its name.

Point Controls

There are four sets of Point controls, corresponding to the four Point In controls in the Controls tab. Uncheck the Show Point checkbox to hide the corresponding Point In control and its crosshair in the viewer. Similarly, edit the Name for Point text field to change the control's name.



The Custom Tool Channel tab

Custom Tool Channels Tab

RGBA, Z, UV Expressions, and XYZ Normal Expressions

The Channel tab is used to set up one expression per each available channel of the image. Each expression is evaluated once per pixel. The result creates the value for that pixel in the output of the image.

Color Channel expressions (RGBA) should generally return floating-point values between 0.0 and 1.0. Values beyond this are clipped if the destination image is an integer. Other expression fields should produce values appropriate to their channel (e.g., between -1.0 and 1.0 for Vector and Normal fields, 0.0 to 1.0 for Coverage, or any value for Depth). The Channel expressions may use the results from both the Setup expressions (as variables s1–s4) and Inter expressions (as variables i1–i4).

Custom Tool Node Syntax

Value Variables	
n1..n8	Numeric Inputs
p1x..p4x	Position Values (X-axis)
p1y..p4y	Position Values (Y-axis)
s1..s4	Setup Expression Results
i1..i4	Inter Expression Results
time	Current Frame
x	Horizontal co-ordinate of the current pixel, between 0.0 and 1.0
y	Vertical co-ordinate of the current pixel, between 0.0 and 1.0
w (or w1..w3)	Width of Image (for image1..image3)
h (or h1..h3)	Height of Image (for image1..image3)

Value Variables

ax (or ax1..ax3)	Image Aspect X (for image1..image3)
ay (or ay1..ay3)	Image Aspect Y (for image1..image3)

NOTE: Use w and h and ax and ay without a following number to get the dimensions and aspect of the primary image.

Channel (Pixel) Variables

c1..c3	Current Channel (for image1..image3)
r1..r3	Red (for image1..image3)
g1..g3	Green (for image1..image3)
b1..b3	Blue (for image1..image3)
a1..a3	Alpha (for image1..image3)
z1..z3	Z-Buffer (for image1..image3)
cv1..cv3	Z Coverage (for image1..image3)
u1..u3	U Coordinate (for image1..image3)
v1..v3 nx1..nx3	V Coordinate (for image1..image3) X Normal (for image1..image3)
ny1..ny3	Y Normal (for image1..image3)
nz1..nz3	Z Normal (for image1..image3)
bgr1..bgr3	Background Red (for image1..image3)
bgg1..bgg3	Background Green (for image1..image3)
bgb1..bgb3	Background Blue (for image1..image3)
bga1..bga3	Background Alpha (for image1..image3)
vx1..vx3	X Vector (for image1..image3)
vy1..vy3	Y Vector (for image1..image3)
nz1..nz3	Z Normal (for image1..image3)

NOTE: Use c1, c2, c3 to refer to the value of a pixel in the current channel. This makes copying and pasting expressions easier. For example, if c1/2 is typed as the red expression, the result would be half the value of the red pixel from image 1, but if the expression is copied to the blue channel, now it would have the value of the pixel from the blue channel.

To refer to the red value of the current pixel in input 1, type `r1`. For the image in input 2, it would be `r2`.

- `get[ch][#]b(x, y)` Read pixel at x,y, or 0 if out of bounds—e.g., `getr1b(0,0)`
- `get[ch][#]d(x, y)` Read pixel at x,y or edge pixel if out of bounds—e.g., `getr1d(0,0)`
- `get[ch][#]w(x, y)` Read pixel at x,y or wrap if out of bounds—e.g., `getr1w(0,0)`

NOTE: There are a variety of methods used to refer to pixels from locations other than the current one in an image.

In the above description, `[ch]` is a letter representing the channel to access. The `[#]` is a number representing the input image. So to get the red component of the current pixel (equivalent to `r`), you would use `getr1b(x,y)`. To get the alpha component of the pixel at the center of image 2, you would use `geta2b(0.5, 0.5)`.

- `getr1b(x,y)` Output the red value of the pixel at position x, y, if there were a valid pixel present. It would output 0.0 if the position were beyond the boundaries of the image (all channels).
- `getr1d(x,y)` Output the red value of the pixel at position x, y. If the position specified were outside of the boundaries of the image, the result would be from the outer edge of the image (RGBA only).
- `getr1w(x,y)` Output the red value of the pixel at position x, y. If the position specified were outside of the boundaries of the image, the x and y coordinates would wrap around to the other side of the image and continue from there (RGBA only).

To access other channel values with these functions, substitute the `r` in the above examples with the correct channel variable (`r`, `g`, `b`, `a` and, for the `getr1b()` functions only, `z`, and so on), as shown above. Substitute the 1 with either 2 or 3 in the above examples to access the images from the other image inputs.

Mathematical Expressions

<code>pi</code>	The value of pi
<code>e</code>	The value of e
<code>log(x)</code>	The base-10 log of x
<code>ln(x)</code>	The natural (base-e) log of x
<code>sin(x)</code>	The sine of x (x is degrees)
<code>cos(x)</code>	The cosine of x (x is degrees)
<code>tan(x)</code>	The tangent of x (x is degrees)
<code>asin(x)</code>	The arcsine of x, in degrees

Mathematical Expressions

<code>acos(x)</code>	The arccosine of x, in degrees
<code>atan(x)</code>	The arctangent of x, in degrees
<code>atan2(x,y)</code>	The arctangent of x,y, in degrees
<code>abs(x)</code>	The absolute (positive) value of x
<code>int(x)</code>	The integer (whole) value of x
<code>frac(x)</code>	The fractional value of x
<code>sqrt(x)</code>	The Square Root of x
<code>rand(x,y)</code>	A random value between x and y
<code>rands(x,y,s)</code>	A random value between x and y, based on seed s
<code>min(x,y)</code>	The minimum (lowest) of x and y
<code>max(x,y)</code>	The maximum (highest) of x and y
<code>dist(x1,y1,x2,y2)</code>	The distance between point x1,y2 and x2,y2
<code>dist3d(x1,y1,z1,x2,y2,z2)</code>	The distance between 3D points x1,y2,z1 and x2,y2,z2
<code>noise(x)</code>	A smoothly varying Perlin noise value based on x
<code>noise2(x, y)</code>	A smoothly varying Perlin noise value based on x and y
<code>noise3(x, y, z)</code>	A smoothly varying Perlin noise value based on x, y and z
<code>if(c, x, y)</code>	returns x if c not 0, otherwise y

Mathematical Operators

<code>!x</code>	1.0 if x = 0, otherwise 0.0
<code>-x</code>	(0.0 - x)
<code>+x</code>	(0.0 + x) i.e. effectively does nothing
<code>x ^ y</code>	x raised to the power of y
<code>x * y</code>	x multiplied by y
<code>x / y</code>	x divided by y
<code>x % y</code>	x modulo y, i.e. remainder of (x divided by y)
<code>x + y</code>	x plus y
<code>x - y</code>	x minus y
<code>x < y</code>	1.0 if x is less than y, otherwise 0.0
<code>x > y</code>	1.0 if x is greater than y, otherwise 0.0

Mathematical Operators

<code>x <= y</code>	1.0 if x is less than or equal to y, otherwise 0.0
<code>x >= y</code>	1.0 if x is greater than or equal to y, otherwise 0.0
<code>x = y</code>	1.0 if x is exactly equal to y, otherwise 0.0
<code>x == y</code>	1.0 if x is exactly equal to y, otherwise 0.0, identical to above
<code>x <> y</code>	1.0 if x is not equal to y, otherwise 0.0
<code>x != y</code>	1.0 if x is not equal to y, otherwise 0.0, i.e. identical to above
<code>x & y</code>	1.0 if both x and y are not 0.0, otherwise 0.0
<code>x && y</code>	1.0 if both x and y are not 0.0, otherwise 0.0, i.e. identical to above
<code>x y</code>	1.0 if either x or y (or both) are not 0.0, otherwise 0.0
<code>x y</code>	1.0 if either x or y (or both) are not 0.0, otherwise 0.0

Examples

The following examples are intended to help you understand the various components of the Custom Tool node.

Rotation

To rotate an image, we need the standard equations for 2D rotation:

$$x' = x * \cos(\theta) - y * \sin(\theta) \quad y' = x * \sin(\theta) + y * \cos(\theta)$$

Using the `n1` slider for the angle θ , and a sample function, we get (for the red channel):

```
getrlb(x * cos(n1) - y * sin(n1), x * sin(n1) + y * cos(n1))
```

This calculates the current pixel's (x,y) position rotated around the origin at (0,0) (the bottom-left corner), and then fetches the red component from the source pixel at this rotated position. For centered rotation, we need to subtract 0.5 from our x and y coordinates before we rotate them, and add 0.5 back to them afterward:

```
getrlb((x-.5) * cos(n1) - (y-.5) * sin(n1) + .5, (x-.5) * sin(n1) + (y-.5) * cos(n1) + .5)
```

Which brings us to the next lesson: Setup and Intermediate Expressions. These are useful for speeding things up by minimizing the work that gets done in the Channel expressions. The Setup expressions are executed only once, and their results don't change for any pixel, so you can use these for `s1` and `s2`, respectively.

```
cos(n1) sin(n1)
```

Intermediate expressions are executed once for each pixel, so you can use these for `i1` and `i2`:

```
(x-.5) * s1 - (y-.5) * s2 + .5
```

```
(x-.5) * s2 + (y-.5) * s1 + .5
```

These are the x and y parameters for the `getrlb()` function from above, but with the Setup results, `s1` and `s2`, substituted so that the trig functions are executed only once per frame, not every pixel. Now you can use these intermediate results in your Channel expressions:

```
getrlb(i1, i2)
```

```
getglb(i1, i2)
```

```
getblb(i1, i2)
```

```
getalb(i1, i2)
```

With the Intermediate expressions substituted in, we only have to do all the additions, subtractions, and multiplications once per pixel, instead of four times. As a rule of thumb, if it doesn't change, do it only once.

This is a simple rotation that doesn't take into account the image aspect at all. It is left as an exercise for you to include this (sorry). Another improvement could be to allow rotation around points other than the center.

Filtering

Our second example duplicates the functionality of a 3 x 3 Custom Filter node set to average the current pixel together with the eight pixels surrounding it. To duplicate it with a Custom Tool node, add a Custom Tool node to the node tree, and enter the following expressions into the Setup tab.

(Leave the node disconnected to prevent it from updating until we are ready.)

S1

```
1.0/w1
```

S2

```
1.0/h1
```

These two expressions are evaluated at the beginning of each frame. `S1` divides 1.0 by the current width of the frame, and `S2` divides 1.0 by the height. This provides a floating-point value between 0.0 and 1.0 that represents the distance from the current pixel to the next pixel along each axis.

Now enter the following expression into the first text control of the Channel tab (r).

```
(getrlw(x-s1, y-s2) + getrlw(x, y-s2) + getrlw(x+s1, y-s2) + getrlw(x+s1,
y) + getrlw(x-s1, y) + r1 +getrlw(x-s1, y+s2) + getrlw(x, y+s2) +
getrlw(x+s1, y+s2)) / 9
```

This expression adds together the nine pixels above the current pixel by calling the `getrlw()` function nine times and providing it with values relative to the current position. Note that we referred to the pixels by using `x+s1`, `y+s2`, rather than using `x+1`, `y+1`.

Fusion refers to pixels as floating-point values between 0.0 and 1.0, which is why we created the expressions we used in the Setup tab. If we had used `x+1`, `y+1` instead, the expression would have sampled the same pixel over and over again. (The function we used wraps the pixel position around the image if the offset values are out of range.)

That took care of the red channel; now use the following expressions for the green, blue, and alpha channels.

```
(getglw(x-s1, y-s2) + getglw(x, y-s2) + getglw(x+s1, y-s2) + getglw(x+s1,
y) + getglw(x-s1, y) + g1 +getglw(x-s1, y+s2) + getglw(x, y+s2) +
getglw(x+s1, y+s2)) / 9
```

```
(getblw(x-s1, y-s2) + getblw(x, y-s2) + getblw(x+s1, y-s2) + getblw(x+s1,
y) + getblw(x-s1, y) + b1 +getblw(x-s1, y+s2) + getblw(x, y+s2) +
getblw(x+s1, y+s2)) / 9
```

```
(getalw(x-s1, y-s2) + getalw(x, y-s2) + getalw(x+s1, y-s2) + getalw(x+s1,
y) + getalw(x-s1, y) + a1 + getalw(x-s1, y+s2) + getalw(x, y+s2) +
getalw(x+s1, y+s2)) / 9
```

It's time to view the results. Add a Background node set to a solid color and change the color to a pure red. Add a hard-edged Rectangular effects mask and connect it to the expression just created.

For comparison, add a Custom Filter node and duplicate the settings from the image above. Connect a pipe to this node from the background to the node and view the results. Alternate between viewing the Custom Tool node and the Custom Filter while zoomed in close to the top corners of the effects mask.

Of course, the Custom Filter node renders a lot faster than the Custom Tool node we created, but the flexibility of the Custom Tool node is its primary advantage. For example, you could use an image connected to input 2 to control the median applied to input one by changing all instances of `getr1w`, `getg1w`, and `getb1w` in the expression to `getr2w`, `getg2w`, and `getb2w`, but leaving the `r1`, `g1`, and `b1s` as they are.

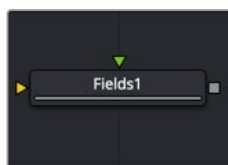
This is just one example; the possibilities of the Custom Tool node are limitless.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Fields [FLDs]



The Fields node

Fields Node Introduction

The Fields node is a robust multipurpose utility offering several functions related to interlaced video frames. It interpolates video fields into frames or video frames into fields. Although the interlace preference and method type is defined in the MediaIn or Loader, and generators, this node can be used to assist in the standards conversion of PAL to NTSC and provides the ability to process fields and frames for specific portions of a node tree.

This node can also interlace two separate images together into a single interlace image. The background input is the dominate field 1 and the foreground is field 2.

Inputs

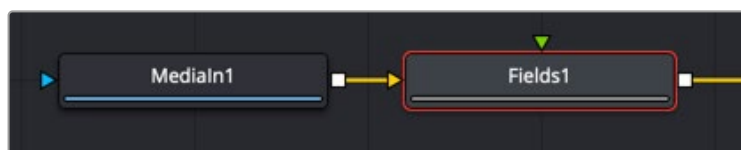
The single input on the Fields node is used to connect a 2D image and an effect mask, which can be used to limit the blurred area.

Stream1 Input: The orange background input is used for the primary 2D image that is interpolated or converted.

Stream2 Input: The optional green foreground input is only used when merging two interlaced images together.

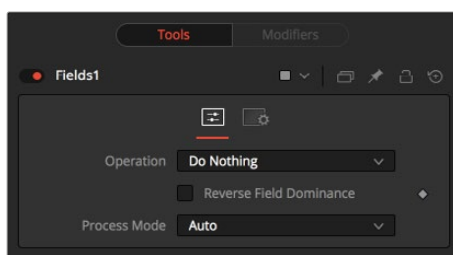
Basic Node Setup

The Fields node below is used to convert the background image from a PAL interlaced format to progressive frames.



The background image can be modified to various interlaced formats.

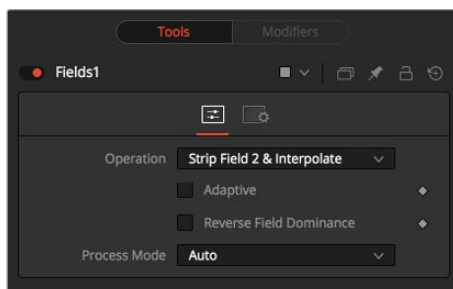
Inspector



The Fields Controls tab

Controls Tab

The Controls tab includes two menus. The Operation menu is used to select the type of field conversion performed. The Process Mode menu is used to select the field's format for the output image.



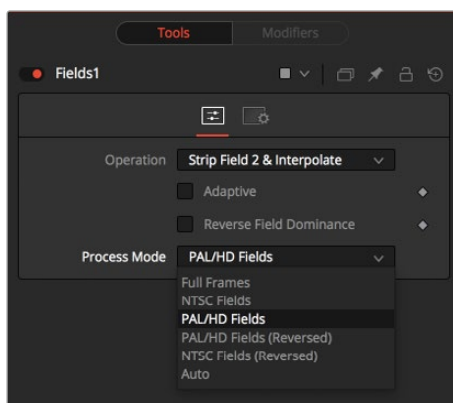
Operation Controls

Operation Menu

- **Do Nothing:** This causes the images to be affected by the Process Mode selection exclusively.
- **Strip Field 2:** This removes field 2 from the input image stream, which shortens the image to half of the original height.
- **Strip Field 1:** This removes field 1 from the input image stream, which shortens the image to half of the original height.
- **Strip Field 2 and Interpolate:** This removes field 2 from the input image stream and inserts a field interpolated from field 1 so that image height is maintained. Should be supplied with frames, not fields.
- **Strip Field 1 and Interpolate:** This removes field 1 from the input image stream and inserts a field interpolated from field 2 so that image height is maintained. Should be supplied with frames, not fields.
- **Interlace:** This combines fields from the input image stream(s). If supplied with one image stream, each pair of frames are combined to form half of the number of double-height frames. If supplied with two image streams, single frames from each stream are combined to form double-height images.
- **De-Interlace:** This separates fields from one input image stream. This will produce double the amount of half-height frames.

Reverse Field Dominance

When selected, the Field Order or Dominance of the image will be swapped.



Process Mode

Process Mode Menu

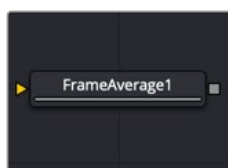
- **Full Frames:** This forces Frame Processing. Useful for processing frames in a part of a node tree that is otherwise field processing.
- **NTSC Fields:** This forces NTSC Field Processing. Useful for processing fields in a part of a node tree that is otherwise frame processing.
- **PAL Fields:** This forces PAL Field Processing. Useful for processing fields in a part of a node tree that is otherwise frame processing.
- **PAL Fields (Reversed):** This forces PAL-swapped Field Processing.
- **NTSC Fields (Reversed):** This forces NTSC-swapped Field Processing.
- **Auto:** This attempts to match the mode of its input images. Fields are used if the input types are mixed.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Frame Average [Avg]



The Frame Average node

Frame Average Node Introduction

The Frame Average node averages together a series of frames to simulate clips shot with long shutter speeds. Aside from motion blur-style effects, it can be useful for time warps or noise removal.

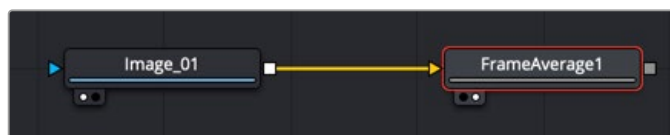
Inputs

The single input on the Frame Average node is used to connect a 2D image that will have the averaging applied.

Input: The orange input is used for the primary 2D image that will be averaged.

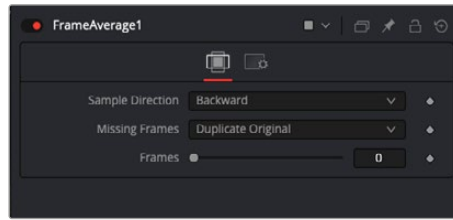
Basic Node Setup

The image connected to the orange input is frame averaged based on the settings in the Inspector.



A Frame Average node blends the input image's frames.

Inspector



The Frame Average Controls tab

Controls Tab

The Controls tab contains the parameters for setting the duration and guidance of the averaged frames.

Sample Direction

The Sample Direction menu determines if the averaged frames are taken before the current frame, after, or a mix of the two.

- **Forward:** Averages the number of frames set by the Frames slider after the current frame.
- **Both:** Averages the number of frames set by the Frames slider, taking frames before and after the current frame.
- **Backward:** Averages the number of frames set by the Frames slider before the current frame.

Missing Frames

This control determines the behavior if a frame is missing from the clip.

- **Duplicate Original:** Uses the last original frame until a new frame is available.
- **Blank Frame:** Leaves missing frames blank.

Frames

This slider sets the number of frames that are averaged.

Keyframe Stretcher [KfS]



The Keyframe Stretcher node

Keyframe Stretcher Node Introduction

The Keyframe Stretcher node is inserted after animated nodes, so the keyframes stretch and the comp's duration is modified. It is used to scale the keyframes on the animation curve to the current duration of the clip. This is particularly useful when creating title templates in Fusion for use in DaVinci Resolve's Edit or Cut page.

TIP: The Keyframe Stretcher can be used on a single parameter by applying the Keystretcher modifier.

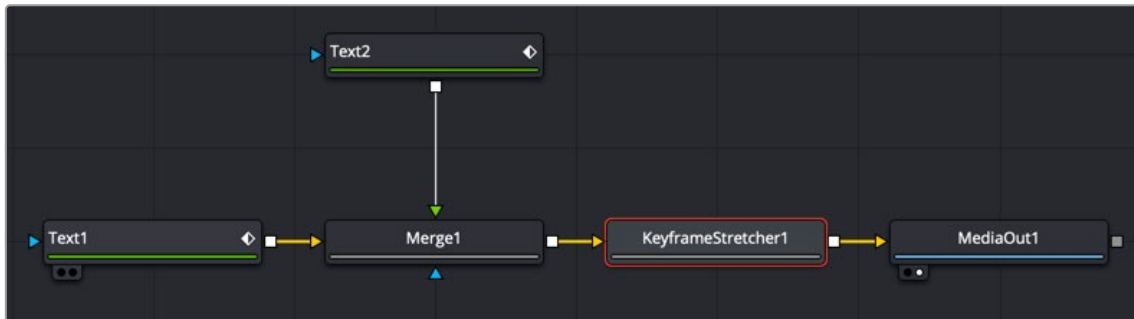
Inputs

The single input on the Keyframe Stretcher node is used to connect a 2D image that contains keyframe animation.

Input: The orange input is used for any node with keyframed animation. The input can be a Merge node that is not animated but contains foreground and background nodes that are animated.

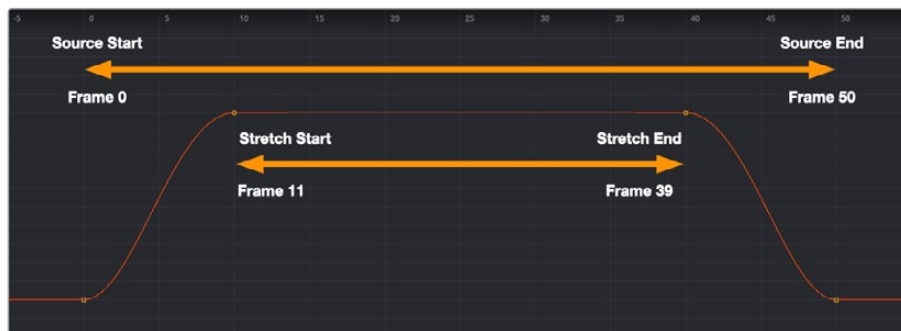
Basic Node Setup

The Keyframe Stretcher is added just before the Media Out or Saver node. All nodes that include animation before the Keyframe Stretcher are modified if the comp changes duration.



A Keyframe Stretcher changing the animation of two text nodes.

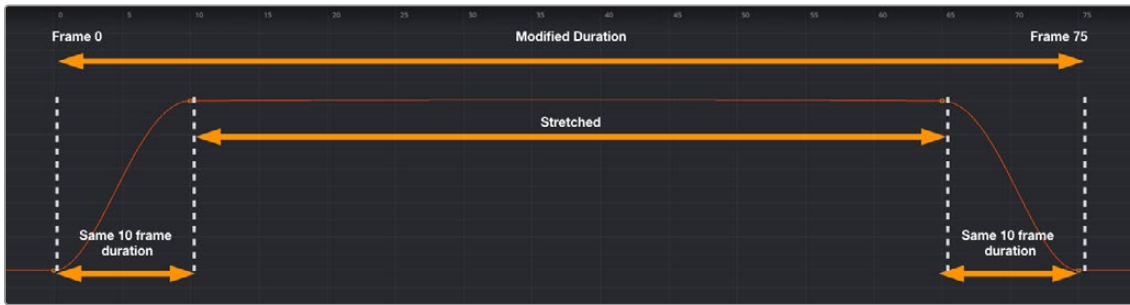
The diagram below shows the original 50-frame animation added to a parameter. The Keyframe Stretcher Start and End would be set to 0 and 50. The second keyframe is set at frame 10, and the third keyframe is set at frame 40. Setting the Stretch Start to frame 11 and the Stretch End to frame 39 will keep the existing keyframes at the same speed (number of frames.) The middle will be stretched.



Original 50-frame animation

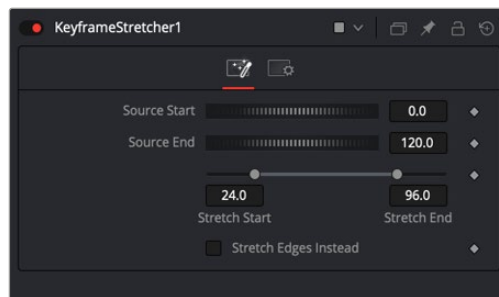
In the below example, the duration of the clip is extended to 75 frames. The first 10 frames and the last 10 frames of the animation run at the same speed as the original animation, while any animation in the middle is stretched to fill the difference.

NOTE: The actual Spline Editor will show only the original keyframe positions. The splines are not changed by the Keyframe Stretcher; only the animation is changed.



Animation modified to 75 frames but stretching only the middle of the animation

Inspector



The Keyframe Stretcher Keyframes tab

Keyframes Tab

The Keyframes tab includes Source controls for setting the source duration and Stretch controls for setting the area of the animation that gets modified.

Source Start/Source End

A source range is specified using the Source Start and Source End controls. These are typically set to match the full range of the animation spline on the Keyframes control.

Stretch Start/Stretch End

The Stretch Start and Stretch End controls let you specify a middle zone where keyframes will be stretched or squished. Handles outside the range will not get scaled. Any keyframes outside the Stretch Start and End range always remain the same number of frames from the Start and End.

Any keyframe adjustments to the original control will be correspondingly scaled back to the source curve and will match the original timing as expected.

Stretch Edges Instead

Enabling the Stretch Edges Instead checkbox overrides the Stretch Start and Stretch End controls and stretches the edges of the animation.

Run Command [Run]



The Run Command node

Run Command Node Introduction

The Run Command node is used to execute an external command or batch file at specific points during a render. You can choose to run a command at the start or the end of a render. Or you can have the command execute once for each frame.

The Run Command can be used to net render other command line applications using the Fusion Render Manager, as well as a host of other useful functions.

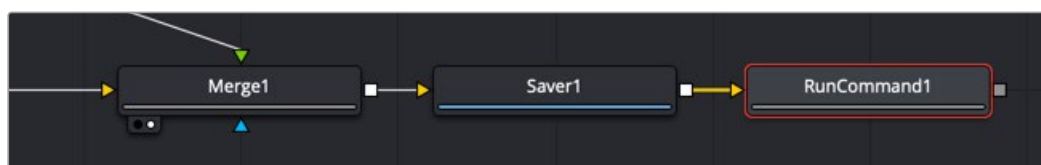
Inputs

The single input on the Run Command node is used to pass through a 2D image.

Input: The optional orange image input is not required for this node to operate. However, if it is connected to a node's output, the Run Command will only launch after the connected node has finished rendering. This is often useful when connected to a Saver, to ensure that the output frame has been fully saved to disk first. If the application launched returns a non-zero result, the node will also fail.

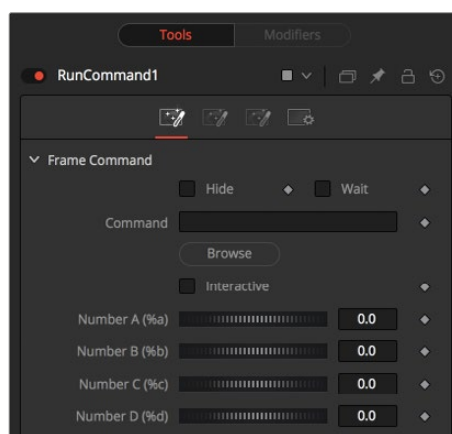
Basic Node Setup

The Run Command node can be connected after a Saver and run once the final frame is completed.



Run Command placed after a Saver node.

Inspector



The Run Command Frame tab

Frame Tab

The Frame tab is where the command to execute is selected and modified.

Hide

Enable the Hide checkbox to prevent the application or script from displaying a window when it is executed.

Wait

Enable this checkbox to cause the node to wait for a remote application or tool to exit before continuing. If this checkbox is disabled, the Fusion continues rendering without waiting for the external application.

Frame Command

This field is used to specify the path for the command to be run after each frame is rendered. The Browse button can be used to identify the path.

Interactive

This checkbox determines whether the launched application should run interactively, allowing user input.

Number A (%a) and Number B (%b)

Various wildcards can be used with the frame commands; these wildcards will be substituted at render time with the correct values.

- **%a:** Outputs the number from the Number A thumbwheel control.
- **%b:** Outputs the number from the Number B thumbwheel control.
- **%t:** Outputs the current frame number (without zero padding).
- **%s:** Substitutes using the text from the large text entry field.

If you want to add zero paddings to the numbers generated by %t, refer to the wildcard with %0x, where x is the number of characters with which to pad the value. This also works for %a and %b.

For example, test%04t.tga would return the following values at render time:

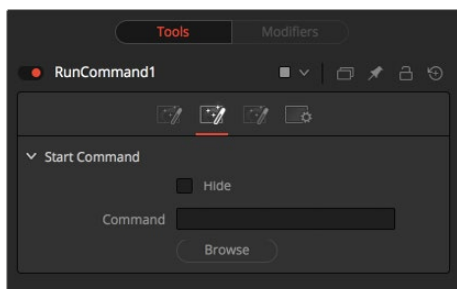
test0000.tga

test0001.tga

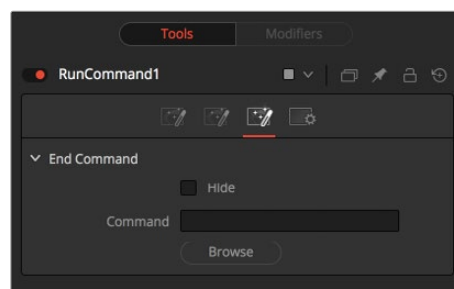
test0009.tga

test0010.tga

You may also pad a value with spaces by calling the wildcard as %x, where x is the number of spaces with which you would like to pad the value.



The Run Command Start tab



The Run Command End tab

Start and End Tabs

The Start and End tabs contain a file browser for a command to be run when the composition starts to render and when the composition is done rendering.

Example

To copy the saved files from a render to another directory as each frame is rendered, save the following text in a file called `copyfile.bat` to your `C\` directory (the root folder).

```
@echo off
set parm=%1 %2
copy %1 %2 set parm=
```

Create or load any node tree that contains a Saver. The following example assumes a Saver is set to output `D\ test0000.exr`, `test0001.exr`, etc. You may have to modify the example to match.

Add a Run Command node after the Saver to ensure the Saver has finished saving first. Now enter the following text into the Run Command node's Frame Command text box:

```
C\copytest.bat D\test%04f.exr C\
```

Select the Hide Frame command checkbox to prevent the Command Prompt window from appearing briefly after every frame.

When this node tree is rendered, each file will be immediately copied to the `C\` directory as it is rendered.

The Run Command node could be used to transfer the files via FTP to a remote drive on the network, to print out each frame as it is rendered, or to execute a custom image-processing tool.

The Run Command node is not restricted to executing simple batch files. FusionScript, VBScript, Jscript, CGI, and Perl files could also be used, as just a few examples.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Set Domain [DOD]



The Set Domain node

Set Domain Node Introduction

Set Domain is used to adjust or set the active area of an image or the area of the image considered to have valid data.

It does not change the image's physical dimensions. Downstream nodes will not process anything outside the Domain of Definition (DoD), thus speeding up rendering of computation-intensive nodes.

This node provides an absolute mode, for setting the domain of definition manually, and a relative mode for adjusting the existing domain of definition.

Inputs

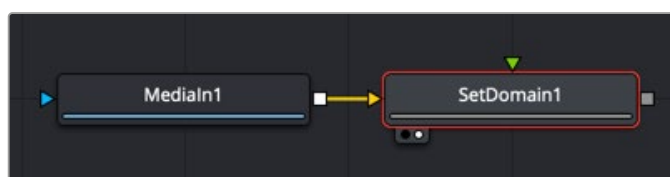
The two inputs on the Set Domain node are used to connect 2D images.

Input: The orange background input must be connected. It accepts a 2D image with the DoD you want to replace or adjust.

Foreground: The green image input is optional but also accepts a 2D image as its input. When the foreground input is connected, the Set Domain node will replace the Background input's domain of definition with the foreground's DoD.

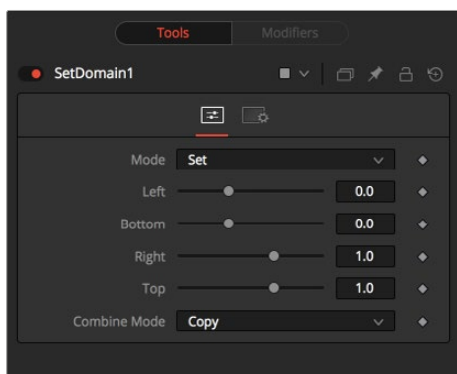
Basic Node Setup

The example below assumes an image is connected to a Set Domain node to manually configure the DoD.

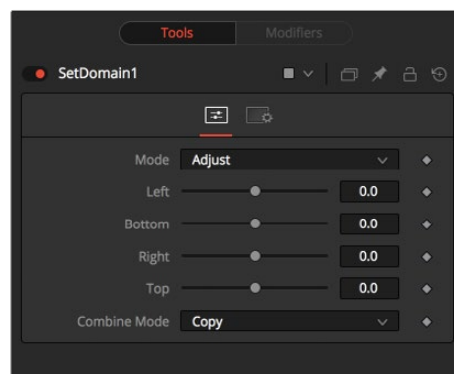


A Set Domain node manually sets the area to limit image processing.

Inspector



The Set Domain Controls tab in Set mode



The Set Domain Controls tab in Adjust mode

Controls Tab

Mode

The Mode menu has two choices depending on whether you want to adjust or offset the existing domain or set precise values for it.

The same operations can be performed in Set or in Adjust mode. In Adjust mode, the sliders default to 0, marking their respective full extent of the image. Positive values shrink the DoD while negative values expand the DoD to include more data.

Set mode defaults to the full extent of the visible image. Sliders default to a scale of 0-1 from left to right and bottom to top.

Left

Defines the left border of the DoD. Higher values on this slider move the left border toward the right, excluding more data from the left margin.

1 represents the right border of the image; 0 represents the left border. The slider defaults to 0 (left border).

Bottom

Defines the bottom border of the DoD. Higher values on this slider move the bottom border toward the top, excluding more data from the bottom margin.

1 represents the top border of the image; 0 represents the bottom border. The slider defaults to 0 (bottom border).

Right

Defines the right border of the DoD. Higher values on this slider move the right border toward the left, excluding more data from the right margin.

1 represents the right border of the image; 0 represents the left border. In Set mode, the slider defaults to 1 (right border).

Top

Defines the top border of the DoD. Higher values on this slider move the top border toward the bottom, excluding more data from the top margin.

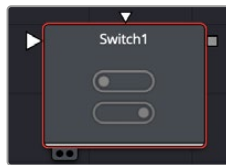
1 represents the top border of the image; 0 represents the bottom border. In Set mode, the slider defaults to 1 (top border).

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Switch [Swi]



The Switch node

Switch Node Introduction

Switch is a tool that enables artists to alternate between multiple input sources, allowing the selection of one output from the chosen inputs. This capability is especially useful for toggling between different visual elements within a composition. The number of input connections can vary, with the node supporting a dynamic number of inputs that can be renamed as needed. Controls in the Config tab allow users to add or remove inputs and change their names.

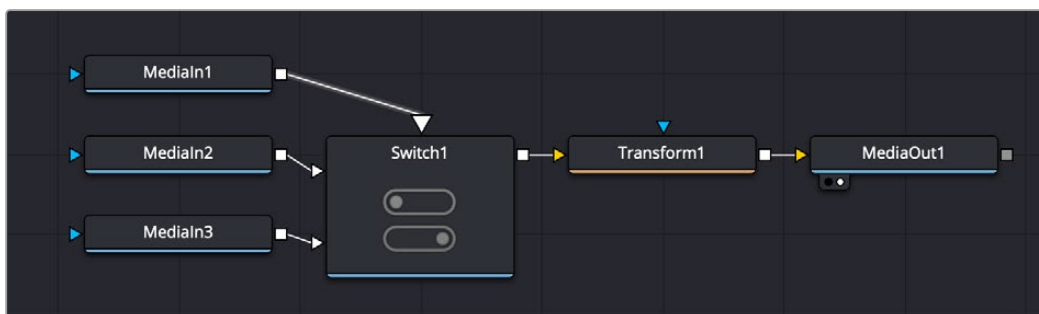
Switch works with most types of tools, including Shapes and 3D. Additionally, users can find the Switch modifier in the Modify With and Insert submenus of a control’s context menu, allowing it to be applied to any supported control.

Inputs

You can add up to nine inputs to a Switch Node using the slider in the Config tools. If you need more than nine sources, you can type that number manually into the Number of Inputs field.

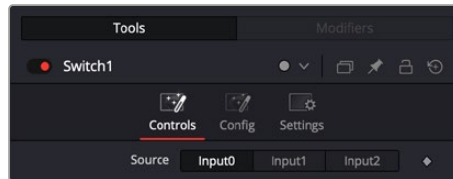
Basic Node Setup

Connect as many input sources as you wish to the Switch Node, and connect its output to the rest of the node tree.



The Switch tool receiving three media inputs, allowing you to choose one to go through the transform node

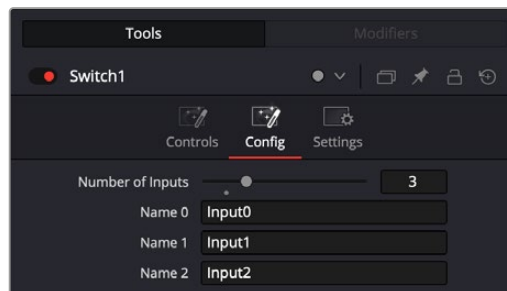
Inspector



The Switch Controls let you choose a source to pass through the switch.

Controls

The Switch Controls consist of a Source switcher that lets you choose which source to pass through the switch.



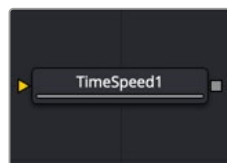
The Switch Configuration

Config

The Switch Configuration lets you choose the number of inputs and rename them.

- **Number of Inputs:** Adjust the slider to add the number of inputs you wish to use, up to nine. If you want more than nine inputs, you can click in the number field and type the number manually.
- **Name X:** You can change the name of the input from numerical to something more descriptive.

Time Speed [TSpd]



The Time Speed node

Time Speed Node Introduction

The Time Speed node allows image sequences to be sped up, slowed down, reversed, or delayed. Image Interpolation offers smooth, high-quality results. Time Speed should be used for static speed changes or to introduce delays in the footage. To apply animated changes in time, such as accelerating or decelerating time, use a Time Stretcher instead.

When operating in Flow mode, Optical Flow data is required. This node does not generate optical flow directly. You have to create it upstream using an Optical Flow node or by loading the forward/reverse vector channels from the image.

TimeSpeed does not interpolate the aux channels but instead destroys them. In particular, the Vector/BackVector channels are consumed and destroyed after computation.

Add an Optical Flow after the Time Speed node if you want to generate flow vectors for the retimed footage.

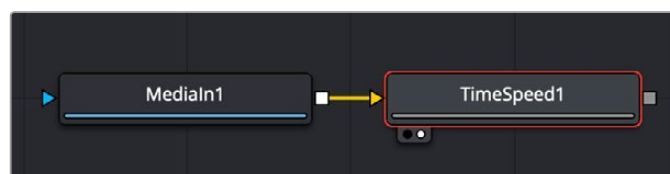
Inputs

The single input on the Time Speed node is used to connect a 2D image that will be retimed.

Input: The orange input is used for the primary 2D image that will be retimed.

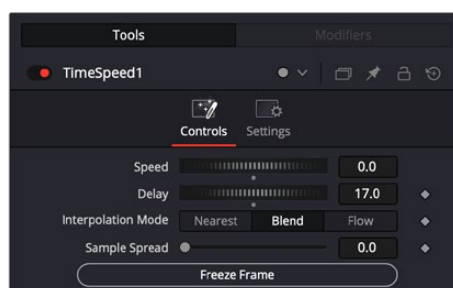
Basic Node Setup

The Time Speed node setup is as simple as connecting a 2D image into the orange background input of the node.



A MediaIn node having its speed changed in the Time Speed node.

Inspector



The Time Speed Controls tab

Speed

This control is used to adjust the Speed, in percentage values, of the outgoing image sequence. Negative values reverse the image sequence. 200% Speed is represented by a value of 2.0, 100% by 1.0, 50% by 0.5, and 10% by 0.1.

The Speed control cannot be animated.

Delay

Use this control to Delay the outgoing image sequence by the specified number of frames. Negative numbers offset time back, and positive numbers advance time.

Interpolate Mode

This menu determines how the time speed is processed in order to improve its visual playback quality, especially in the case of clips that are slowed down. There are three choices in the menu.

- **Nearest:** The most processor efficient and least sophisticated method of processing; frames are either dropped for fast motion or duplicated for slow motion.
- **Blend:** Also processor efficient, but can produce smoother results; adjacent duplicated frames are dissolved together to smooth out slow or fast motion effects.
- **Flow:** The most processor intensive but highest quality method of speed effect processing. Using vector channels pre-generated from an Optical Flow node, new frames are generated to create slow or fast motion effects. The result can be exceptionally smooth when motion in a clip is linear. However, two moving elements crossing in different directions or unpredictable camera movement can cause unwanted artifacts.

Sample Spread

This slider is displayed only when Interpolation is set to Blend. The slider controls the strength of the interpolated frames on the current frame. A value of 0.5 blends 50% of the frame before and 50% of the frame ahead and 0% of the current frame.

Depth Ordering

This menu is displayed only when Interpolation is set to Flow. The Depth Ordering is used to determine which parts of the image should be rendered on top. This is best explained by example.

In a locked-off camera shot where a car is moving through the frame, the background does not move, so it produces small, or slow, vectors. The car produces larger, or faster, vectors.

The Depth Ordering, in this case, is Fastest on Top, since the car draws over the background.

In a shot where the camera pans to follow the car, the background has faster vectors, and the car has slower vectors, so the Depth ordering method would be Slowest on Top.

Clamp Edges

This checkbox is displayed only when Interpolation is set to Flow. Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges can cause a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use clamp edges only to correct small gaps around the edges of an interpolated frame.

Edge Softness

This slider is only displayed when Interpolation is set to Flow and Clamp Edges is enabled. It helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

These checkboxes are displayed only when Interpolation is set to Flow. These controls determine which frames and which vectors are used to create the in-between frames. Each method ticked on will be blended into the result.

- **Prev Forward:** Takes the previous frame and uses the Forward vector to interpolate the new frame.
- **Next Forward:** Takes the next frame in the sequence and uses the Forward vector to interpolate the new frame.
- **Prev Backward:** Takes the previous frame and uses the Back Forward vector to interpolate the new frame.
- **Next Backward:** Takes the next frame in the sequence and uses the Back vector to interpolate the new frame.

Freeze Frame

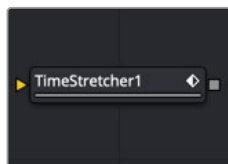
Pressing this button automatically adjusts the Speed and Delay controls to freeze playback on the currently selected frame.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Time Stretcher [TSt]



The Time Stretcher node

Time Stretcher Node Introduction

The Time Stretcher node is similar to the Time Speed node but permits the speed of the clip to be animated. Full spline control of the effect is provided, including smoothing. As a result, the Time Stretcher can be used to animate a single clip to 200, back to normal speed, pause for a second, and then play backward (like a VCR rewinding).

Image interpolation offers smooth, high-quality results, all using a spline curve to adjust time nonlinearly. To apply constant time changes such as frame rate changes, use a Time Speed instead.

When operating in Flow mode, Optical Flow data is required. This node does not generate optical flow directly; you must create it manually upstream using an Optical Flow node or by loading the forward/reverse vector channels from disk.

Flow Stretcher does not interpolate the aux channels but instead destroys them. In particular, the Vector/BackVector channels are consumed/destroyed. Add an Optical Flow after the Flow Stretcher if you want to generate flow vectors for the retimed footage.

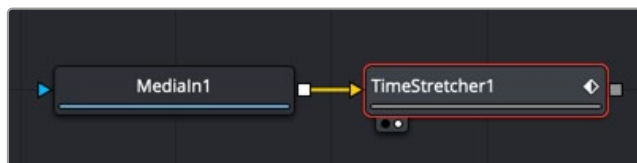
Inputs

The single input on the Time Stretcher node is used to connect a 2D image that will be time stretched.

Input: The orange input is used for the primary 2D image that will be time stretched.

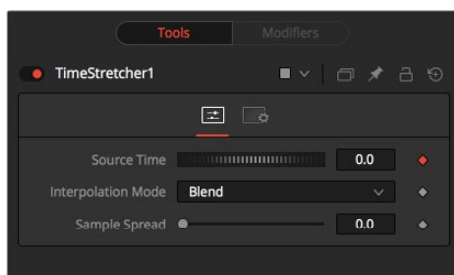
Basic Node Setup

Like the Time Speed node, the Time Stretcher setup is as simple as connecting a 2D image into the orange background input of the node.



A MediaIn node having its time ramped to various speeds in the Time Stretcher node.

Inspector



The Time Stretcher controls

Source Time

This control designates from which frame in the original sequence to begin sampling.

When a Time Stretcher node is added to the node tree, the Source Time control already contains a Bézier spline with a single keyframe set to 0.0. The keyframe position is determined by the current time when the node is added to the node tree.

NOTE: The Source Time spline may not be immediately visible until Edit is selected from the Source Time's contextual menu, or Display all Splines is selected from the Spline Window's contextual menu.

Interpolate Mode

This menu determines the how the time speed is processed in order to improve its visual playback quality, especially in the case of clips that are slowed down.

There are three choices in the menu.

- **Nearest:** The most processor efficient and least sophisticated method of processing; frames are either dropped for fast motion or duplicated for slow motion.
- **Blend:** Also processor efficient but can produce smoother results; adjacent duplicated frames are dissolved together to smooth out slow or fast motion effects.

- **Flow:** The most processor intensive but highest quality method of speed effect processing. Using vector channels pre-generated from an Optical Flow node, new frames are generated to create slow or fast motion effects. The result can be exceptionally smooth when motion in a clip is linear. However, two moving elements crossing in different directions or unpredictable camera movement can cause unwanted artifacts.

Sample Spread

This slider is displayed only when Interpolation is set to Blend. The slider controls the strength of the interpolated frames on the current frame. A value of 0.5 blends 50% of the frame before and 50% of the frame ahead and 0% of the current frame.

Depth Ordering

This menu is displayed only when Interpolation is set to Flow. The Depth Ordering is used to determine which parts of the image should be rendered on top. This is best explained by example.

In a locked-off camera shot where a car is moving through the frame, the background does not move, so it produces small, or slow, vectors. The car produces larger, or faster, vectors.

The Depth Ordering in this case is Fastest on Top, since the car draws over the background.

In a shot where the camera pans to follow the car, the background has faster vectors, and the car has slower vectors, so the Depth ordering method would be Slowest on Top.

Clamp Edges

This checkbox is displayed only when Interpolation is set to Flow. Under certain circumstances, this option can remove the transparent gaps that may appear on the edges of interpolated frames. Clamp Edges can cause a stretching artifact near the edges of the frame that is especially visible with objects moving through it or when the camera is moving.

Because of these artifacts, it is a good idea to use clamp edges only to correct small gaps around the edges of an interpolated frame.

Edge Softness

This slider is displayed only when Interpolation is set to Flow and Clamp Edges is enabled. It helps to reduce the stretchy artifacts that might be introduced by Clamp Edges.

If you have more than one of the Source Frame and Warp Direction checkboxes turned on, this can lead to doubling up of the stretching effect near the edges. In this case, you'll want to keep the softness rather small at around 0.01. If you have only one checkbox enabled, you can use a larger softness at around 0.03.

Source Frame and Warp Direction

These checkboxes are displayed only when Interpolation is set to Flow. These controls determine which frames and which vectors are used to create the in-between frames. Each method ticked on will be blended into the result.

- **Prev Forward:** Takes the previous frame and uses the Forward vector to interpolate the new frame.
- **Next Forward:** Takes the next frame in the sequence and uses the Forward vector to interpolate the new frame.
- **Prev Backward:** Takes the previous frame and uses the Back Forward vector to interpolate the new frame.
- **Next Backward:** Takes the next frame in the sequence and uses the Back vector to interpolate the new frame.

Example

Make sure that the current time is either the first or last frame of the clip to be affected in the project. Add the Time Stretcher node to the node tree. This will create a single point on the Source Time spline at the current frame. The value of the Source Time will be set to zero for the entire Global Range.

Set the value of the Source Time to the frame number to be displayed from the original source, at the frame in time it will be displayed in during the project.

To shrink a 100-frame sequence to 25 frames, follow these steps:

1. Change the Current Time to frame 0.
2. Change the Source Time control to 0.0.
3. Advance to frame 24.
4. Change the Source Time to 99.
5. Check that the spline result is linear.
6. Fusion will render 25 frames by interpolating down the 100 frames to a length of 25.
7. Hold the last frame for 30 frames, and then play the clip backward at regular speed. Continue the example from above and follow the steps below.
8. Advance to frame 129.
9. Right-click on the Source Time control and select Set Key from the menu.
10. Advance to frame 229 (129 + 100).
11. Set the Source time to 0.0.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Wireless Link [Wire]



The Wireless Link node

Wireless Link Node Introduction

The Wireless Link node helps manage the tangle of connection lines in a node tree by wirelessly connecting one 2D node to another 2D node.

Although Wireless Links can be helpful, try to keep as much of a node tree as visible as possible; otherwise, you lose one of the main benefits of a node tree.

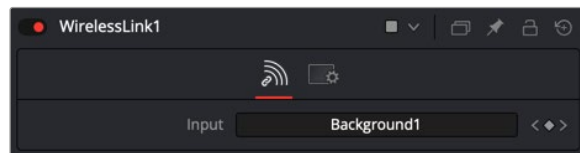
Inputs

There are no inputs on this node.

Basic Node Setup

There is no setup for this node. It is a free-standing node that connects “wirelessly” using the control in the Inspector.

Inspector



The Wireless Link Controls tab

Controls Tab

The Controls tab in the Wireless Link node contains a single Input field for the linked node.

Input

To use the Wireless Link node, in the Node Editor, drag the 2D node into the Input field of the Wireless Link node. Any change you make to the original node is wirelessly replicated in the Wireless Link node. You can use the output from the Wireless Link node to connect to a nearby node.

Common Controls

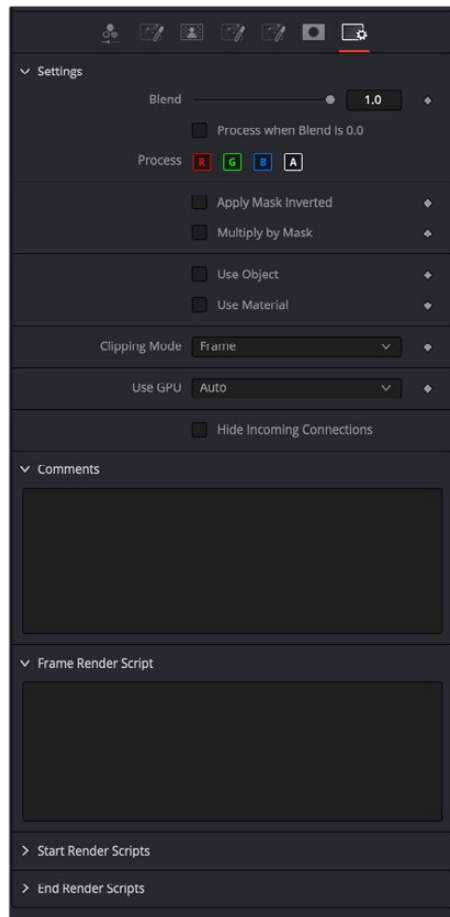
Settings Tab

The Settings tab in the Inspector is also duplicated in other miscellaneous nodes. These common controls are described in detail in the following “The Common Controls” section.

The Common Controls

Nodes that handle miscellaneous operations share a number of identical controls in the Inspector. This section describes controls that are common among miscellaneous nodes.

Inspector



The Common Miscellaneous Settings tab

Settings Tab

The Settings tab in the Inspector can be found on every tool in the miscellaneous nodes. The controls are consistent and work the same way for each tool.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the view. The image or sequence must have been rendered from a 3D software package with those channels included.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.